Arda Korkmaz

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Skills

- **Languages**: Python, SQL, Bash, C/C++, Java
- **ML Skills**: Deep Learning, Computer Vision, NLP, LLMs, RAG, Data Mining
- **ML & Data Libraries**: PyTorch, TensorFlow, transformers, scikit-learn, XGBoost, AutoGluon, Ultralytics (YOLO), OpenCV, spaCy, Stanza, pandas, NumPy
- **LLM Tooling**: LangChain, Ollama, llama.cpp
- **Systems, Backend & DevOps**: Linux, PostgreSQL, SQLite, FastAPI, Docker, Git, Jenkins, CI/CD, Snakemake, QEMU

Education

- **San Jose State University**
  Master of Science in Software Engineering — *Expected 2028*
- **UC San Diego Extended Studies**
  Machine Learning Engineering Bootcamp — *2025*
- **Ozyegin University**
  Bachelor of Science in Computer Science — *2020*
- **Bornova Anadolu Lisesi**
  Highschool Degree — *2014*

Experience

- **Software Engineer** — Istanbul, Turkey
  Massive Bioinformatics — *Oct. 2021 - July 2026*
  - **Clinical Microbiome Agent**: Built a multi-agent LLM clinical decision-support system that gives physicians patient-specific insights by cross-referencing gut microbiome results with phenotype questionnaires, surfacing findings neither source reveals alone. Developed it through an iterative harness of isolated-context agents: claim extraction checked by a deterministic verifier, and a blind 7-dimension judge whose missed cross-references and weaknesses drove each prompt revision. Benchmarked Claude and local 27–31B models, tuning context windows to fit customer GPUs; the local deployment cut hallucinations by 40%.
  - **RAG Pipelines**: Automated manual expert curation of the company's variant knowledge base by building RAG pipelines (custom chunking, FAISS retrieval) to extract structured clinical evidence (e.g., variant–therapy associations) from long, noisy documents, incl. NCCN guidelines across 30+ cancer types.
  - **Pipeline-to-Desktop Framework**: Built a reusable toolkit for packaging bioinformatics pipelines as self-contained Linux/Windows desktop apps. Each app pairs a Flet interface with isolated Docker or QEMU execution and bundles its own binaries and native libraries, so it runs offline on ordinary hardware with live progress monitoring. Used to ship pipelines both as commercial products for customers and as internal tools.
  - **Backend & Microservices**: Built key parts of Massive Analyser's shift from a Django monolith to FastAPI microservices. Most notably, re-architected the platform's core analysis dispatch engine as a standalone microservice that drives the full lifecycle of every pipeline run (launch, monitoring, and termination) across varied pipeline types, inputs, and parameters, with run state tracked in the database.
  - **Analysis Reporting Tools**: Built a Python PDF reporting library with custom page-layout logic, used across pipelines by two teams. Also developed Jinja2-based HTML reports for Massive Analyser pipelines, designing templates around backend data models and developing the backend rendering layer.
  - **Education and Training**: Delivered training on Linux and core CS topics to learners of varying experience levels. Course material included operating systems, computer networks, and shell scripting, tailored to both beginners and intermediate learners.

- **R&D Intern** Istanbul, Turkey
  Netaş, Cloud and Managed Services Software Development and Support Department *Summer 2019*
  - **Proof of Concept**: Conducted research on the applications of blockchain technology for an European Commission funded project, *Critical Chains*. Developed proof of concept projects using Solidity and Hyperledger Fabric to demonstrate feasibility. Presented the findings to senior executives.

- **Software Engineering and R&D Intern** Istanbul, Turkey
  Idea Teknoloji Çözümleri *Summer 2018*
  - **Full Stack .NET**: Worked on an in-house work-hour tracking system using .NET and MySQL. Implemented several features and integrated them with the existing database.
  - **NGS reporting**: Prepared an introductory report on Next Generation Sequencing (NGS), for internal training purposes. Documented methodologies, workflows, and key steps involved in NGS.

Projects

- **Real-Time Food Detection System**
  Developed a real-time food tray detection system using CV to automate restaurant billing by identifying tray contents. Built as the capstone project for UCSD ML Engineering Bootcamp, featuring a complete pipeline from data collection to deployment.
  - **Technical Implementation**: Engineered a dual-stage YOLO detection system with custom-trained models for tray detection and food classification. Implemented Lucas-Kanade optical flow tracking with frame skipping to enable live inference, achieving 3.6x average and up to 6.2x speedup on standard hardware without GPU acceleration. Built a FastAPI WebSocket server with a model registry supporting on-the-fly replacement and configuration, real-time video streaming with multiple client interfaces.
  - **Dataset Creation and Collection**: Collected and curated multiple datasets including Food-101 (101k images) and UECFoodPix (10k segmented images) for research and experimentation. Built custom dataset **TrayTrack** for tray detection. Collected 103 images and expanded to 1,854 through targeted augmentation. Fine-tuned YOLO11n for tray detection. Achieved 95.65% mAP@0.5 on UECFOOD100 experimental models, ultimately selecting robust baseline models for stable real-time performance.
  - **Tech Stack**: Python, PyTorch, Ultralytics YOLO, OpenCV, FastAPI, WebSockets, Shapely, Albumentations, Argon2 auth

Languages

- **Turkish**: Native proficiency
- **English**: Professional working proficiency
- **German**: Limited proficiency